

EAST WEST UNIVERSITY

Department of Computer Science & Engineering

Course: CSE489 — Mobile Programming

PROJECT REPORT

MessManager

A Role-Based Android Application for Bachelor Mess & Shared Accommodation Management

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1. Executive Summary

MessManager is a native Android application engineered to automate and modernize communal living ('Mess') operations commonly practiced by university students and bachelor professionals in South Asia. In conventional setups, mess management relies on physical paper notebooks and manual spreadsheet calculations. This legacy workflow frequently results in arithmetic errors, bitter financial disputes, uncontrollable food wastage, and unfair administrative burdens placed upon individual students.

MessManager resolves these inefficiencies by offering a centralized, cloud-backed platform powered by Google Cloud Firestore and Firebase Authentication, designed under the Model-View-ViewModel (MVVM) architectural pattern. The system introduces an automated Dynamic Meal Rate Engine, Timed Cut-off Meal Locks, Equal Utility Cost Distribution, Selective Special Feast Billing, and an in-app Democratic Manager Rotation System. The application delivers end-to-end transparency, real-time balance calculations, offline data persistence, and automated month-closing balance rollovers.

2. Introduction

2.1 Background & Domain Context

In urban university hubs across Bangladesh, tens of thousands of university students and young working professionals live in shared rental apartments known locally as a 'Mess'. In a mess, residents pool financial resources to cover recurring living expenses:

- **1. Daily Grocery (Bazaar) & Food Costs:** Variable costs of purchasing raw ingredients, vegetables, meat, and fish.
- **2. Fixed Shared Utilities:** Apartment rent, housemaid (Bua) wages, high-speed WiFi, electricity, piped/cylinder gas, water supply, and building garbage fees.
- **3. Special Feasts:** Occasional Friday biryani, holiday barbecues, or festival feasts organized for a subset of residents.

Traditionally, one resident is designated as the 'Manager' for a month, responsible for recording daily meals on wall charts, logging market expenses in paper notebooks, collecting cash deposits, and calculating complex final balance sheets at the end of the month.

2.2 Problem Statement

The conventional manual system suffers from significant operational flaws:

- **Arithmetic Mistakes:** Calculating fractional meal rates and variable utility splits by hand often leads to discrepancies and mistrust.
- **Food Wastage:** Without rigid cut-off times, members turn off meals after groceries have already been purchased and cooked.
- **Manager Burnout:** Managing finances manually creates an excessive burden on individual students, while lack of an audit trail makes dispute resolution impossible.
- **Special Feast Conflicts:** When special feasts are organized, manually tracking who opted in versus who is absent and segregating their costs from standard daily meals is extremely difficult.

- **Balance Carryover Dilemmas:** Manually carrying forward positive and negative debt balances into the subsequent month is tedious and error-prone.

3. Motivation

3.1 Social & Practical Significance

Living in a shared mess is an essential stage of student and bachelor life. However, friction over money, unverified deposits, and missed meals regularly causes tension among roommates. Automating this domain through a native mobile application delivers:

- **Absolute Financial Transparency:** Every member can inspect the live dynamic meal rate, mess fund balance, daily shopping logs, and pending cash deposit approvals in real-time.
- **Food Wastage Reduction:** Timed meal locks enforce strict deadlines before cooking begins.
- **Democratic Fairness:** Periodic elections allow mess members to elect their finance and meal managers democratically.

3.2 Key Challenges in Manual Mess Management

Comparison: Legacy Manual Mess vs. Automated MessManager

1. Paper Ledger Records: Prone to physical damage, loss, or tampering.
2. Unsettled Cash Debts: Members unaware of real-time balance debt.
3. Uncontrolled Cooking: Food cooked for absent members leading to waste.
4. Unfair Responsibility: Same person stuck managing every month.
5. Special Feast Mixing: Non-participating members charged for feasts.

4. Project Objectives & Scope

4.1 Primary Objectives

- **1. Modern Android Architecture:** Build the system using Android Jetpack MVVM, ViewBinding, StateFlow, Coroutines, and Navigation Component.
- **2. Role-Based Access Control (RBAC):** Provide tailored UI dashboards for Super Admins, Finance Managers, Meal Managers, and General Members.
- **3. Dynamic Financial Engine:** Calculate live dynamic meal rates, equal utility shares, and individual carried balances in real-time.
- **4. Meal Lock Cut-off Enforcement:** Prevent food waste by locking meal modifications past designated daily cut-off times.
- **5. Democratic Governance:** Implement nomination, multi-candidate voting, and automated role handover for monthly managers.
- **6. One-Click Month Audit:** Provide an automated 'Close Month' routine that freezes financial summaries and carries net balances forward.

4.2 Core Scope & Deliverables

The project scope encompasses a complete native Android application running on Android 8.0 (API 26) through Android 14 (API 34), fully integrated with Firebase Authentication, Google Cloud Firestore, and Firebase Cloud Storage.

4.3 Role-Based Access Control (RBAC) Matrix

Feature / Permission	Super Admin	Finance Mgr	Meal Mgr	Member
Create Mess / Generate Invite Code	YES	NO	NO	NO
Approve / Reject Member Join Requests	YES	NO	NO	NO
Assign Roles / Kick Members	YES	NO	NO	NO
Configure Mess Settings (Name, Due Limit)	YES	NO	NO	NO
Trigger & Close Manager Elections	YES	NO	NO	NO
Emergency Delete Mess	YES	NO	NO	NO
Log Daily Bazaar (Market) Expenses	NO	YES	NO	NO
Manage Fixed Shared Bills (Rent, WiFi)	NO	YES	NO	NO
Approve / Reject Deposit Requests	NO	YES	NO	NO
Execute 'Close Month' Audit & Rollover	NO	YES	NO	NO
Set Daily Menus & Cut-off Lock Times	NO	NO	YES	NO
Inspect Live Meal Counts & Roster	NO	NO	YES	NO
Create Special Meal Polls (e.g. Biryani)	NO	NO	YES	NO
Toggle Daily Meals ON/OFF (Before Lock)	YES	YES	YES	YES
Submit Deposit Payment Requests	YES	YES	YES	YES
Vote in Elections & Special Meal Polls	YES	YES	YES	YES
Read Notice Board & Edit Profile	YES	YES	YES	YES

**Note: Admins and Managers are active members of the mess who eat daily meals and share expenses. Their elevated roles grant administrative privileges in addition to standard member capabilities.*

5. System Architecture & Methodology

5.1 Architecture Pattern (MVVM & Clean Architecture)

MessManager is built following the official Android Jetpack MVVM (Model-View-ViewModel) architectural pattern. The codebase is organized into three distinct layers:

- **1. Presentation Layer:** Fragments render UI via ViewBinding and observe immutable StateFlow emissions from ViewModels.
- **2. Domain / ViewModel Layer:** ViewModels manage state, process UI events, enforce coroutine lifecycles (viewModelScope), and execute business rules.

- **3. Data Layer:** Repositories abstract data operations, bridge Firebase Tasks using `.await()`, and interface with Cloud Firestore and Firebase Authentication.

5.2 Tech Stack & Android Jetpack Components

- **Language & JVM:** Kotlin 1.9.24 running on Java 11 bytecode.
- **UI Framework:** Single Activity architecture (MainActivity) hosting NavHostFragment and Material Components 1.10.0.
- **Jetpack Components:** Navigation Component (nav_graph.xml), Lifecycle ViewModel & LiveData KTX, ViewBinding.
- **Asynchronous Concurrency:** Kotlinx Coroutines with structured concurrency (viewModelScope, withTimeout, repeatOnLifecycle).
- **Firebase Cloud Platform:** Firebase Auth (Auth & Session), Cloud Firestore (NoSQL Persistent Ledger), Firebase Storage (Profile images).
- **Image Handling:** Glide 4.16.0 with circular cropping and placeholder fallbacks.

5.3 Database Design & Cloud Firestore Schemas

Cloud Firestore contains 12 dedicated NoSQL collections designed for high scalability and zero-composite index requirements:

- **1. messes:** MessId, name, inviteCode, superAdminUid, currentMonthId, dueThresholdBdt, lock times, daily menus.
- **2. users:** Uid, name, email, phone, emergencyContact, messId, role, carried balance, joinApproved flag.
- **3. mealEntries:** Composite ID (`{userId}_{date}_{mealType}`), count (1.0 or 0.0), isGuestMeal.
- **4. bazaarEntries:** EntryId, messId, date, amount, addedByUid, optional linkedPollId for feasts.
- **5. fixedBills:** BillId, messId, monthId, type (Rent, Maid, Wifi, etc.), amount, addedBy.
- **6. deposits:** DepositId, messId, memberUid, amount, date, status ('pending' or 'approved').
- **7. specialMealPolls:** PollId, messId, title, eventDate, optedInUserIds list, closed flag.
- **8. electionPolls:** PollId, messId, title, monthId, options (candidate UIDs), financeVotes map, mealVotes map, status.
- **9. monthSummaries:** SummaryId, messId, monthId, totalStandardBazaar, totalFixedBills, mealRate, memberSummaries list.
- **10. notices:** NoticeId, messId, title, content, postedBy, date timestamp.
- **11. inventoryItems:** ItemId, messId, itemName, isLowStock boolean flag.
- **12. bazaarRoster:** RosterId, messId, date, assignedMemberUid, assignedMemberName.

5.4 Mathematical & Financial Calculation Engine

Core Financial Formulas Implemented in MessManager

1. Dynamic Meal Rate (R_{meal}):

$$R_{\text{meal}} = (\text{Total Standard Bazaar Expenses}) / (\text{Total Standard Meals Consumed in Month})$$

*Special meal bazaar expenses linked to polls are excluded from R_{meal} .

2. Member's Monthly Meal Cost (C_{meal}):

$$C_{meal,i} = (\text{Total Meals Consumed by Member } i) \times R_{meal}$$

3. Equal Fixed Utility Bill Share (S_{fixed}):

$$S_{fixed} = (\text{Total Shared Bills: Rent + Maid + WiFi + Gas + Electricity}) / (\text{Approved Members})$$

4. Dedicated Special Meal Cost Share ($S_{special}$):

$$S_{special,i} = \text{sum}(\text{Special Meal Cost}_p / \text{Opted-in Members Count}_p) \text{ for all polls opted-in by member } i$$

5. Live Net Member Balance (B_i):

$$B_i = \text{OpeningBalance}_i + \text{ApprovedDeposits}_i - (C_{meal,i} + S_{fixed} + S_{special,i})$$

6. Detailed System Modules & Workflows

6.1 Authentication, Session Management & Onboarding

The authentication flow handles email/password registration and sign-in. Upon successful login, the session interceptor in AuthViewModel checks:

- **No Mess Joined:** If user.messId is null, the user is navigated to MessSetupFragment.
- **Unapproved Member:** If user.joinApproved is false, the user is navigated to PendingApprovalFragment.
- **Active Member:** If approved, the user is routed directly to their designated role dashboard via navigateToRoleDashboard().

6.2 Super Admin Administrative Module

The Super Admin (creator of the mess) governs the household:

- **Manage Members:** Reviews pending join requests with one-tap Approve/Reject controls. Removes/kicks existing members.
- **Role Assignment:** Promotes members to Finance Manager or Meal Manager roles.
- **Mess Settings:** Customizes due threshold limits (e.g. 500 BDT) and enables Ramadan mode.
- **Emergency Teardown:** Performs complete mess deletion with batch disassociation of all members.

6.3 Finance Management & Expense Auditing Module

The Finance Manager manages the mess treasury:

- **Daily Bazaar Logging:** Records daily shopping expenses. Optionally tags an expense to an open Special Meal poll.
- **Fixed Costs & Utilities:** Maintains recurring apartment rent, maid salary, internet bills, and utility expenses.

- **Deposit Verification:** Reviews pending deposit payment requests submitted by members, verifying cash/MFS receipt.
- **Financial Overview:** Live inspection of total mess reserves, expense breakdowns, and individual member debt balances.

6.4 Meal Management, Menu & Cut-off Lock System

The Meal Management module is a core pillar of MessManager, designed to eliminate daily cooking miscalculations and food waste:

- **Universal Demand Aggregation:** The Meal Manager dashboard aggregates live Breakfast, Lunch, and Dinner demand counts across the entire mess in real time.
- **Meal Attendance Inspection:** Provides a detailed roster showing exactly which members are eating today's meals, enabling accurate portion planning.
- **Automated Cut-off Lock Enforcement:** The Meal Manager configures cut-off lock times (e.g., 07:00 for breakfast, 12:30 for lunch, 20:00 for dinner). When a member attempts to toggle a meal switch, the ViewModel validates `LocalTime.now()` against the configured deadline. If past the deadline, the toggle is blocked, ensuring food is cooked strictly for confirmed eaters.
- **Daily Menu Broadcasting:** The Meal Manager publishes daily cooking menus (e.g., 'Khichuri & Egg Curry') which are displayed directly on every resident's home dashboard.
- **Ramadan Mode:** When toggled in settings, the system seamlessly transitions meal categories from Breakfast/Lunch/Dinner to Sehri/Iftar/Dinner with custom cut-off lock schedules.

6.5 General Member Daily Operations

General Members interact with the core daily features:

- **Personal Snapshot:** Displays live financial balance, meals eaten today (e.g., 3/3), and current dynamic meal rate.
- **Meal Toggles:** Switches to turn daily meals ON/OFF before the cut-off lock deadline.
- **Deposit Requests:** Sends deposit payment requests directly to the Finance Manager.
- **Red Zone Due Tracker:** Alerts the member when their balance drops below the mess due threshold.

6.6 Special Feast (Biryani/Event) Polling & Dedicated Billing

Special meals (such as Friday Biryani or festival feasts) are managed through dedicated polls. Members vote YES (Opt-in) or NO (Opt-out). When the Finance Manager purchases feast ingredients, the expense is linked to the poll ID, ensuring that the cost is billed strictly to participating members and completely bypassed from the regular daily meal rate.

6.7 Democratic Manager Election & Rotation System

To avoid manager fatigue and ensure shared responsibility, the Super Admin triggers monthly manager elections. All approved members are nominated as candidates for Finance Manager and Meal Manager. Members vote confidentially in-app, with live vote share calculations.

6.8 End-of-Month Audit & Automatic Balance Rollover

When the month concludes, the Finance Manager executes 'Close Month':

- **1. Audits Final Totals:** Calculates final dynamic meal rates, utility shares, and individual closing balances.
- **2. Archives Snapshot:** Writes an immutable MonthSummary audit document to Firestore.
- **3. Carries Over Balances:** Sets each user's balance to their closing balance, which becomes next month's opening balance.
- **4. Advances Month:** Advances currentMonthId to the next calendar month (e.g., 2026-08 to 2026-09) and closes active elections.

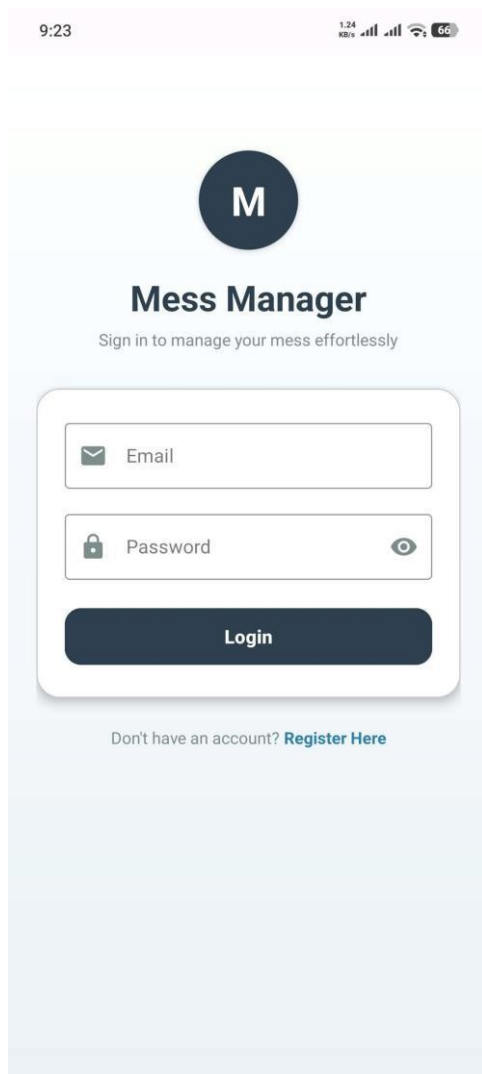
6.9 Digital Notice Board & Shared Inventory

- **• Notice Board:** Broadcasts announcements posted by managers and admins with author names and timestamps.
- **• Pantry Inventory:** Tracks shared supplies (Rice, Oil, Gas Cylinder) with a one-tap 'Low Stock' flag alerting upcoming shoppers.

7. User Interface (UI) & Screen Walkthroughs

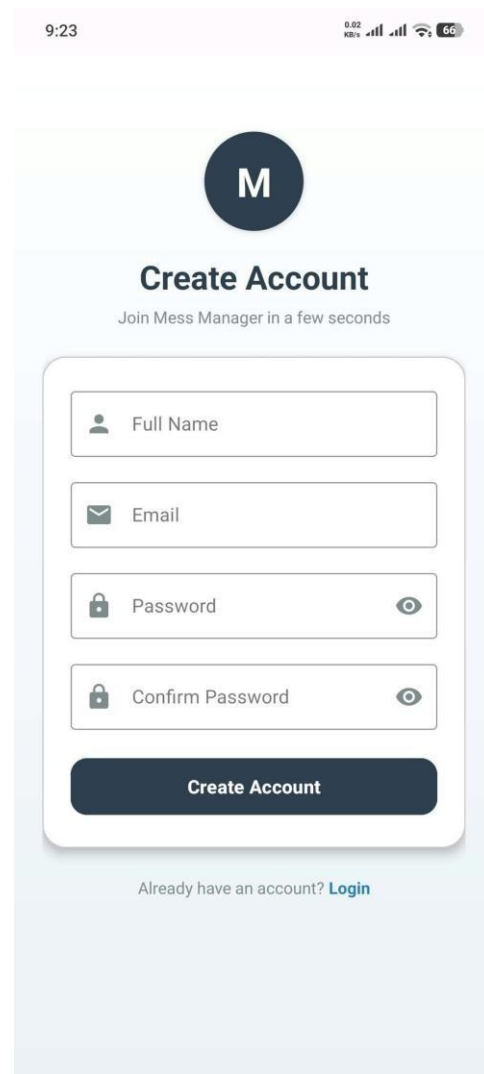
The following section presents the implemented user interface screens captured directly from the running Android application.

7.1 Authentication & Onboarding Screens



The screen displays a mobile interface for 'Mess Manager'. At the top, a status bar shows the time 9:23, signal strength, and 6G connectivity. Below the status bar is a dark blue circle with a white 'M'. The title 'Mess Manager' is centered, followed by the subtitle 'Sign in to manage your mess effortlessly'. The main form contains two input fields: 'Email' with an envelope icon and 'Password' with a lock icon and a toggle eye icon. A dark blue 'Login' button is positioned below the fields. At the bottom, a link reads 'Don't have an account? [Register Here](#)'.

Figure 7.1: User Sign In



The screen displays a mobile interface for 'Create Account'. At the top, a status bar shows the time 9:23, signal strength, and 6G connectivity. Below the status bar is a dark blue circle with a white 'M'. The title 'Create Account' is centered, followed by the subtitle 'Join Mess Manager in a few seconds'. The main form contains four input fields: 'Full Name' with a person icon, 'Email' with an envelope icon, 'Password' with a lock icon and a toggle eye icon, and 'Confirm Password' with a lock icon and a toggle eye icon. A dark blue 'Create Account' button is positioned below the fields. At the bottom, a link reads 'Already have an account? [Login](#)'.

Figure 7.2: Account Registration

Figures 7.1 & 7.2: Authentication interfaces featuring email/password input, real-time regex form validation, and role session routing.

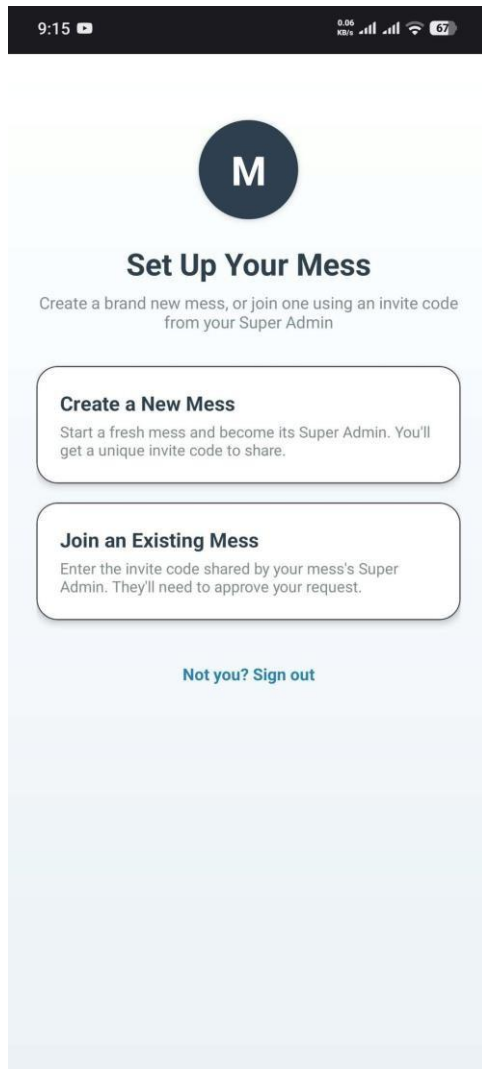


Figure 7.3: Mess Setup & Join Options

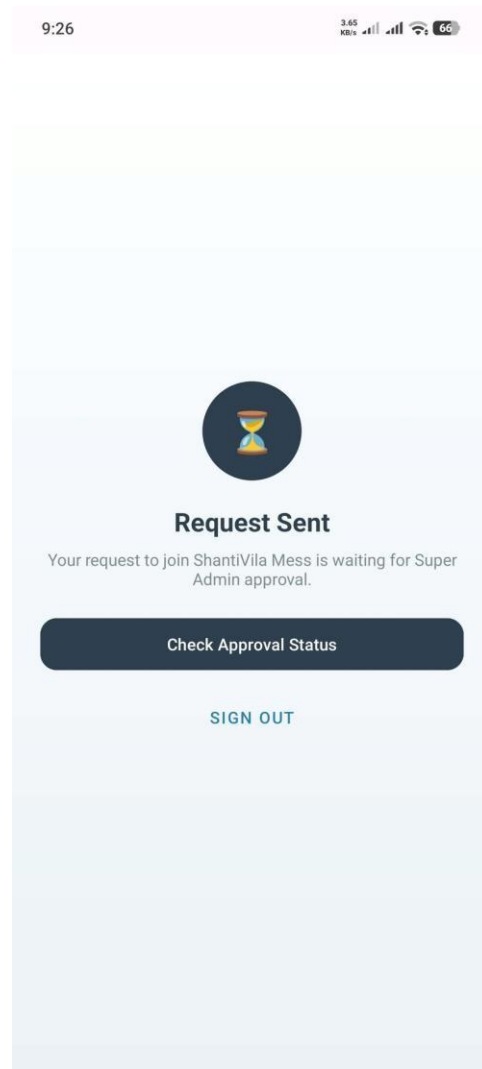


Figure 7.4: Pending Approval Waiting Screen

Figures 7.3 & 7.4: Onboarding workflow allowing users to create a new mess (becoming Super Admin) or enter a 6-digit invite code to await admission.

7.2 Member & Meal Management Screens

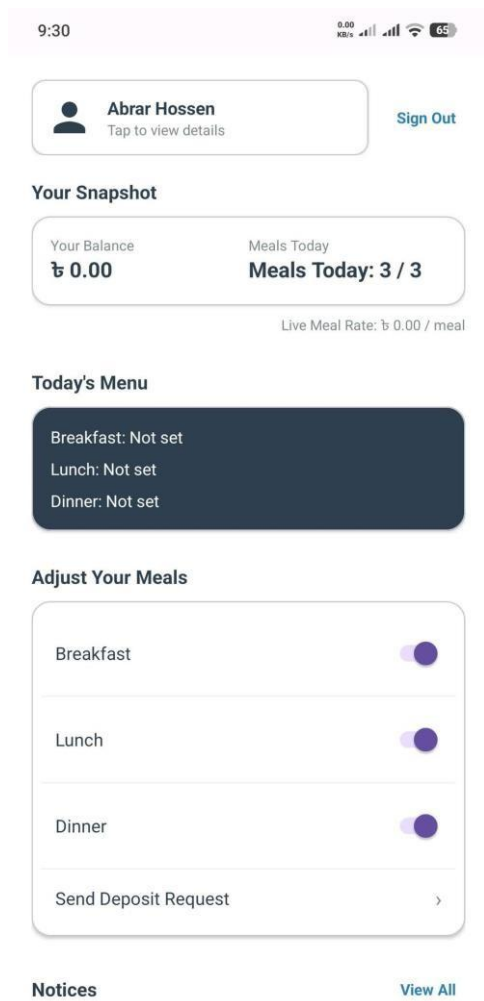


Figure 7.5: General Member Dashboard

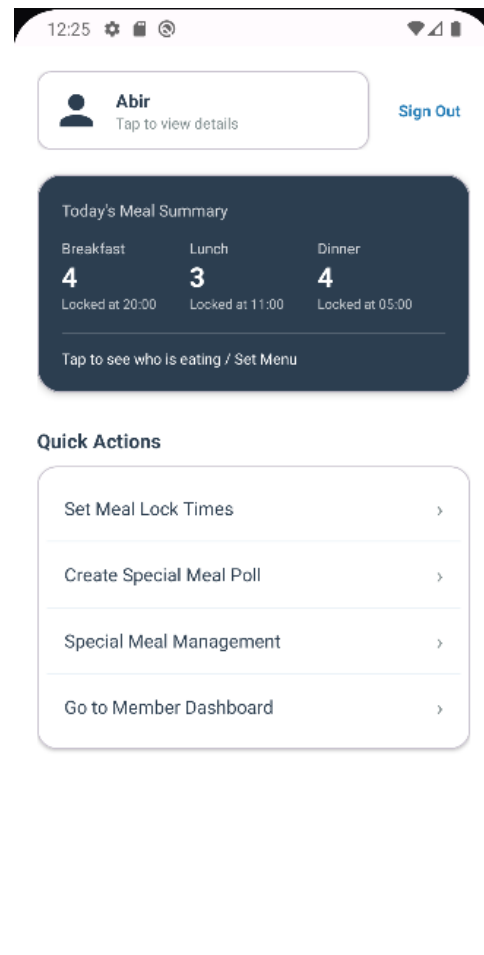


Figure 7.6: Meal Manager Summary & Lock Times

Figures 7.5 & 7.6: (Left) General Member home screen showing balance, meal switches, daily menu, and notices. (Right) Meal Manager dashboard showing universal meal demand counts and lock times.

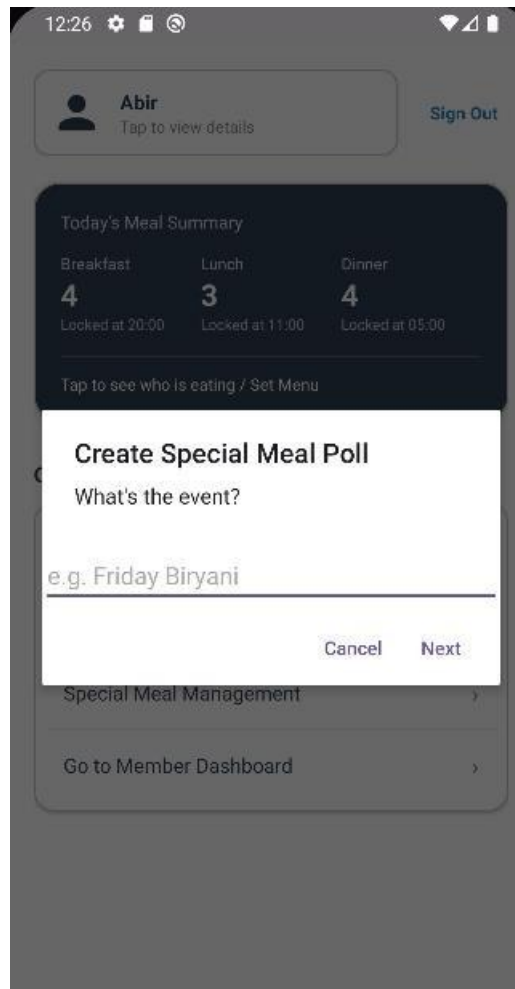


Figure 7.7: Special Feast (Biryani) Poll Creation. Enables Meal Managers to create dedicated feast polls with participant opt-in tracking.

7.3 Super Admin Governance Screens

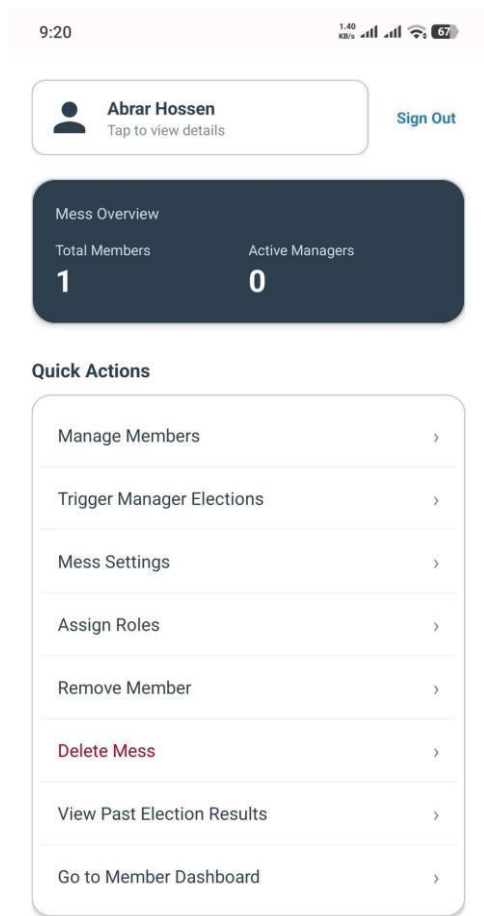


Figure 7.8: Super Admin Overview

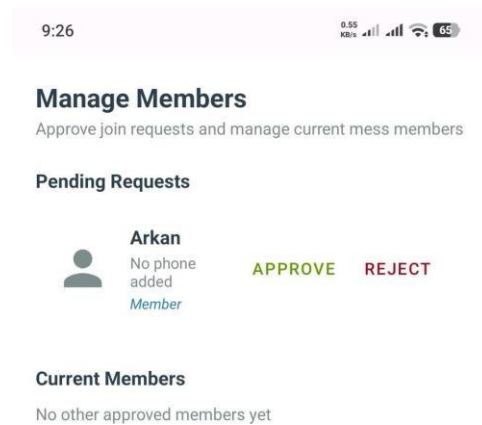
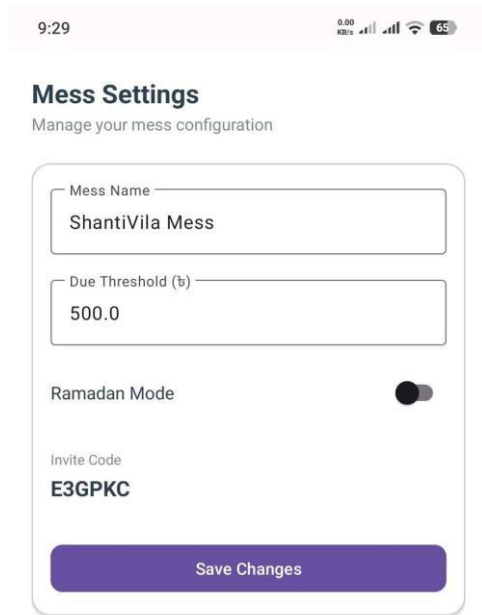


Figure 7.9: Member Join Approval List

Figures 7.8 & 7.9: Administrative interfaces for reviewing active managers, admitting new members, and assigning roles.



9:29 0.00 KB/s 65%

Mess Settings

Manage your mess configuration

Mess Name
ShantiVila Mess

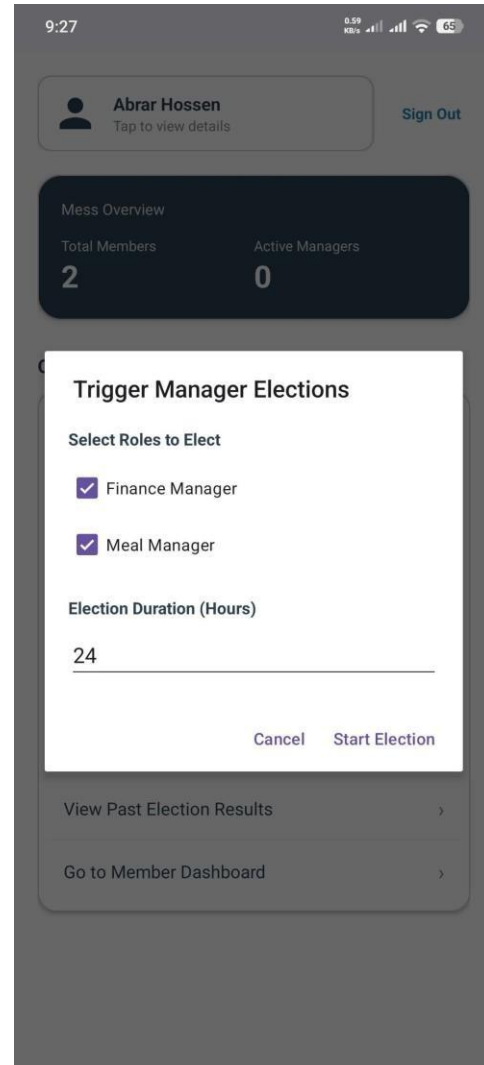
Due Threshold (₹)
500.0

Ramadan Mode ☐

Invite Code
E3GPKC

Save Changes

Figure 7.10: Mess Configuration & Settings



9:27 0.59 KB/s 65%

Abrar Hossen
Tap to view details [Sign Out](#)

Mess Overview

Total Members	Active Managers
2	0

Trigger Manager Elections

Select Roles to Elect

- ☒ Finance Manager
- ☒ Meal Manager

Election Duration (Hours)

24

Cancel Start Election

[View Past Election Results](#)

[Go to Member Dashboard](#)

Figure 7.11: Trigger Election Dialog

Figures 7.10 & 7.11: Settings interface for due threshold limits and Ramadan mode, alongside the manager election trigger dialog.

7.4 Finance Management & Month-End Auditing

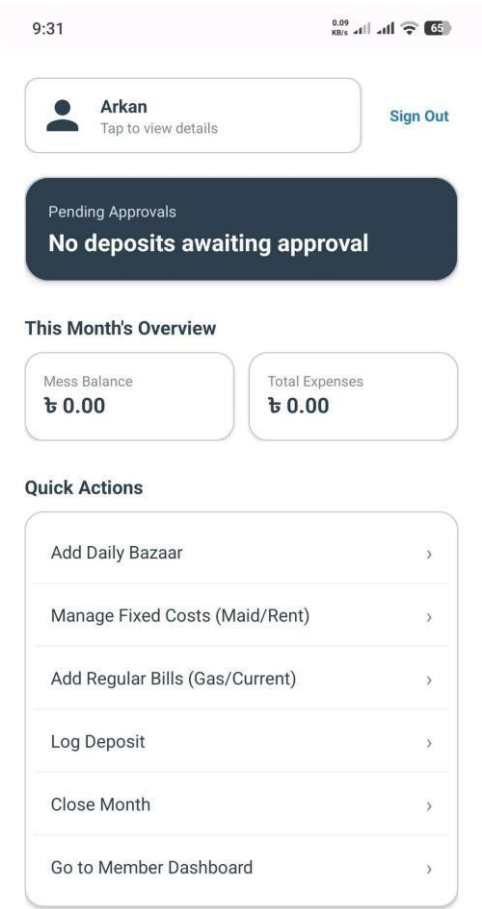


Figure 7.12: Finance Manager Dashboard

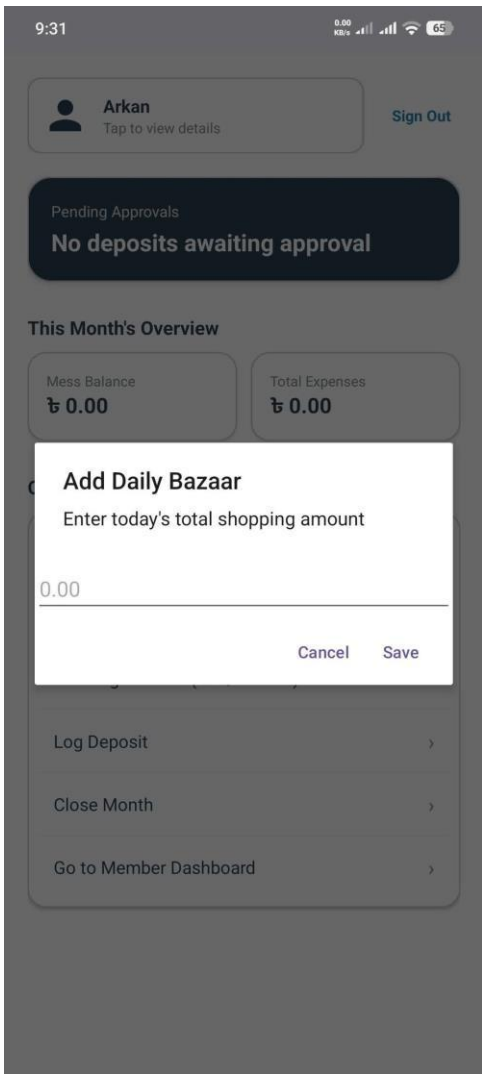


Figure 7.13: Add Daily Bazaar Shopping Entry

Figures 7.12 & 7.13: Finance dashboard displaying pending deposit verifications and the daily grocery market entry interface.

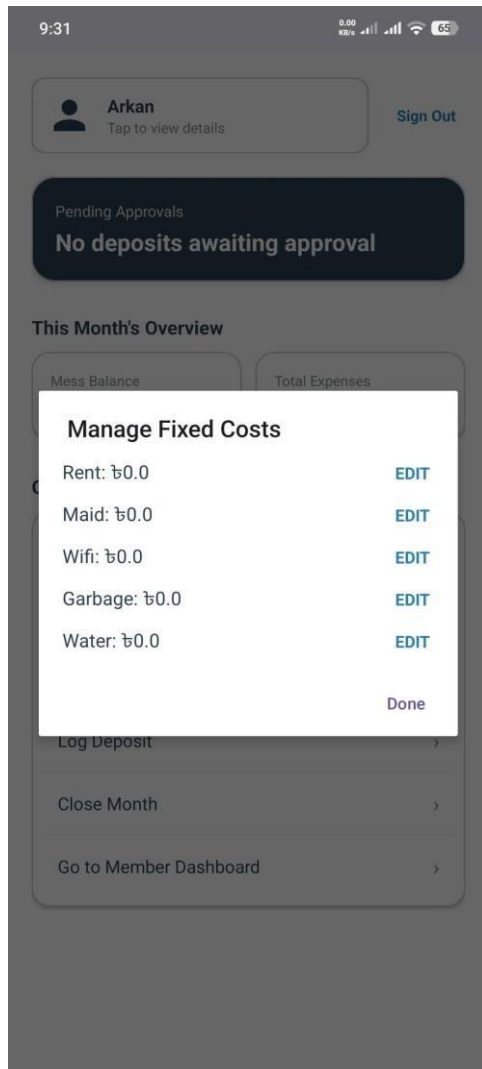


Figure 7.14: Manage Fixed Shared Bills

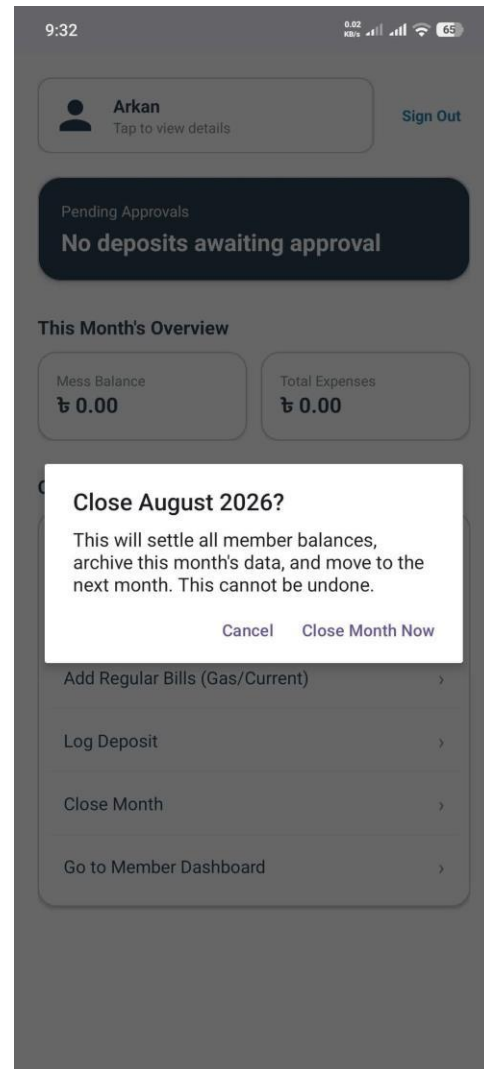


Figure 7.15: Close Month Settlement Dialog

Figures 7.14 & 7.15: Recurring utility bill manager (Rent, Maid, WiFi) and the one-click Close Month settlement routine.

7.5 Democratic Election & User Profile

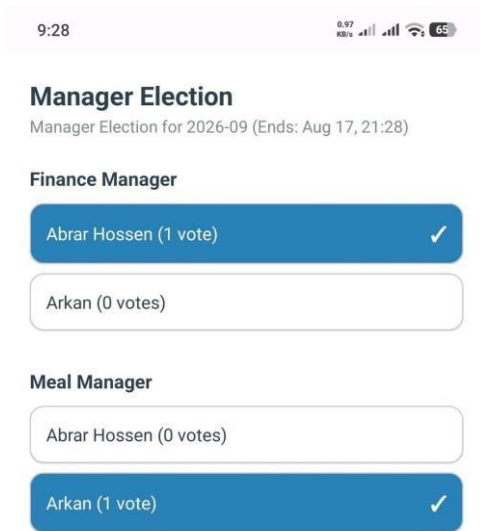


Figure 7.16: Manager Election Ballot Screen

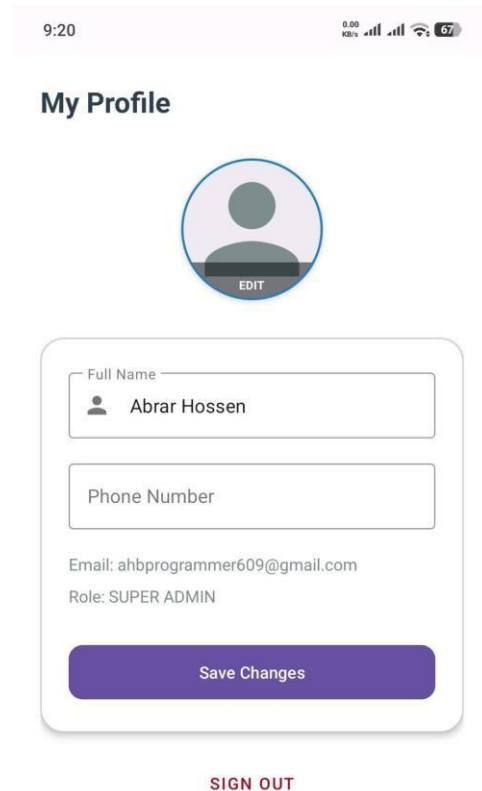


Figure 7.17: User Profile Management

Figures 7.16 & 7.17: Democratic election voting screen with live candidate vote counts, and user profile management.

8. Non-Functional Requirements & Performance Considerations

- **Offline Data Persistence:** Enabled via `PersistentCacheSettings` in `MessManagerApp`. Users can view cached balances, meal statuses, and notices without an active internet connection.
- **Deterministic Document Indexing:** Composite keys (`${userId}_${date}_${mealType}`) ensure atomic updates and prevent duplicate meal entries.
- **Efficient Single-Equality Queries:** Firestore reads use simple equality filters (`whereEqualTo('messId', messId)`), performing complex sorting in-memory to eliminate composite index crashes.

- **Memory Leak Prevention:** ViewBinding references are cleared in `onDestroyView()` and StateFlow emissions are collected strictly within `Lifecycle.State.STARTED`.
- **Mathematical Error Guards:** All rate and balance formulas include division-by-zero guards (if `totalMeals > 0 ... else 0.0`).

9. Limitations & Future Enhancements

9.1 Current Limitations

- **Manual Deposit Verification:** Deposits are transferred via external cash or MFS (bKash/Nagad) and verified manually by the Finance Manager.
- **In-App Notifications:** Reminders currently operate in-app rather than via background Firebase Cloud Messaging (FCM) push services.

9.2 Future Enhancements

- **Payment Gateway Integration:** Automated deposit logging via bKash, Nagad, and SSLCommerz APIs with instant transaction ID reconciliation.
- **Receipt OCR Scanning:** Automatic extraction of shopping amounts from physical bazaar receipts using Google ML Kit Text Recognition.
- **Graphical Analytics:** Visual charts showing monthly spending trends, grocery inflation, and meal consumption histograms using MPAndroidChart.
- **Cloud Function Triggers:** Automated server-side execution of election deadlines and month-closing triggers on the 1st of every month.

10. Conclusion

MessManager successfully addresses the logistical and financial challenges of bachelor mess living by replacing error-prone manual paper diaries with a secure, automated, and role-based mobile system. Built on modern Android architectural standards (MVVM, Clean Architecture, Jetpack Navigation, Coroutines, and Cloud Firestore), the application delivers:

- **Financial Transparency:** Real-time visibility with an automated dynamic meal rate engine.
- **Food Waste Prevention:** Strict enforcement of timed cut-off meal locks and attendance aggregation.
- **Fair Cost Distribution:** Equal utility cost splitting and dedicated special meal event billing.
- **Democratic Accountability:** Transparent monthly manager rotations and automated balance carryover.

The project demonstrates the practical application of mobile application engineering principles to solve an authentic community problem, significantly improving trust, convenience, and financial clarity for residents in shared accommodation.

11. References

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